Technical report — ParliParla

- **Track:** Track 2A — OpenParlData: Extracting Parliamentary Affairs from PDFs into One Common Structure
- **Event:** Online (1–16 October 2026)
- **Team:** ParliParla — Julian Hermes (solo)
- **Repository:** https://github.com/Avonik/ParliParla
- **Demo:** `make run` (CLI; no video required for this challenge)

1. Summary

OpenParlData holds 326,000 parliamentary affairs from 89 Swiss parliaments. Their register metadata is harmonised; their *content* — what a member asked, what the executive answered to which question, what was decided — is locked in PDFs in three languages and dozens of layouts. We designed a small common schema for exactly that content layer (**ParliParla Affair Schema, PAS v0.1**), built an **Apertus v1.5 8B** converter that fills it **with verified provenance** (every value carries a quote that code has located in the source, or it is marked *unsure*), and ran it on a stratified sample of **260 affairs from 18 parliaments** (federal, 11 cantons, 6 cities; DE/FR/IT; motions, postulates, interpellations, written questions).

Headline results: the *intervention* side of the schema — sub-questions and demands with verbatim text, submitters, date, addressed body — is extractable with verified quotes for **92–100 % of affairs in 17 of 18 parliaments**, by an 8B model, for **USD 0.66 per 1,000 affairs**. The *answer* side is a different story, and not because of the model: only 7 of 18 parliaments publish the executive's answer as a document that OpenParlData can attach, and a keyword scan shows the pipeline missed an existing answer in just 2 of 164 unanswered affairs. Where answers exist, Apertus aligns answer parts to sub-questions and classifies the stance (answered / partial / declined / deferred) — which lets us answer, for the first time, *what share of an interpellation's questions the executive actually addresses*, per parliament.

Our feasibility verdict: **a common schema for the intervention layer is feasible today across all three levels of government; the answer and decision layers are feasible only where the parliament's publication practice makes them visible, which is a data-publication gap, not a modelling gap.** Section 8 maps that gap parliament by parliament.

2. The question we answer

The challenge asks whether one machine-readable structure can express a parliamentary affair so that two parliaments become comparable, and whether an LLM can fill it from real PDFs without inventing values. Register-level fields (type, submitter, dates) are already harmonised by OpenParlData and have been studied by other teams. We deliberately target the layer below:

- the *items* of an intervention (numbered questions, demands),
- the *answer parts* of the executive and **which item each part answers**,
- the executive's *recommendation* and the parliament's *decision*,
- and, for every value, *where in which document it comes from*.

This is the layer that makes cross-parliamentary questions answerable ("how often are sub-questions left unanswered?") and that no existing standard (OParl, Popolo, Akoma Ntoso) models explicitly.

3. The representation model: PAS v0.1

`docs/schema.md` documents the schema; `docs/schema.json` is the JSON Schema; `src/parliparla/schema.py` is the Pydantic source of truth. Four design decisions matter:

1. **Every value is a claim with evidence.** All extracted values are a `Fld = {value, status, provenance[], note}` with `status ∈ {extracted, inferred, unsure, not_found, not_applicable}`. `extracted` means the quote was located in the source by code; `inferred` means derived (e.g. from the register); `unsure` means the model proposed it but the quote could not be found. A consumer who wants facts filters on `extracted`.
2. **Provenance is a span.** `{doc_id, char_start, char_end, quote}` into the document text as served by OpenParlData, so a reviewer in the extraction workbench can jump to the passage.
3. **Documents have roles.** An affair is a bundle of documents, and whether a document is the intervention, the executive's response, a combined text, minutes or a decision extract is itself extracted — because that is where parliaments differ most (Section 8).

4. **Small vocabularies, originals kept.** Ten affair types, eight actor roles, five answer stances, eight outcomes; the parliament's own wording is always stored alongside.

Structure (abridged): `Affair{body, type, title, actors[], addressed_to, submitted_date, intervention{preamble, items[{n, label, kind, text}]}, response{author, date, general_statement, items[{n, answers_items[], stance, text}], recommendation}, decision{outcome, date, decided_by, votes}, history[], related[], documents[{doc_id, role, …}], extraction{model, tokens, seconds}}`.

4. The converter

Pipeline per affair (`src/parliparla/extract.py`):

1. **Source documents.** PDF texts from `/affairs/{id}/docs` (PDFBox extraction served by OpenParlData) and, for the Federal Assembly, HTML `texts` converted to plain text with list numbering preserved. Newspaper pages and audio links that some parliaments attach are filtered.
2. **Document roles.** Rules on the document title first (`Antwort`, `réponse`, `Protokoll`, …); when the title is uninformative (`Versione PDF`), Apertus classifies the first 1,800 characters.
3. **Intervention prompt** → title, type, submitters (+ party), date, addressed body, items.
4. **Response prompt** (given the items) → author, date, answer parts with `answers_items` and stance, recommendation. Executed on `executive_response` and `combined` documents.
5. **Decision prompt** on minutes / decision documents → outcome, date, votes.
6. **Verification.** Every quote is searched in the source text: exact after whitespace and hyphenation normalisation, else fuzzy (RapidFuzz partial alignment ≥ 88). Found → `extracted` with offsets; not found → `unsure`. The model cannot assert a value without a quote.

All three prompts are in `src/parliparla/prompts.py` (one JSON object each, verbatim-quote rules, no translation, nulls instead of guesses). JSON mode, temperature 0, seed 42. Everything is OpenAI-compatible, so the same code runs against Public AI, vLLM or llama.cpp.

5. Use of Apertus

- **Model:** `swiss-ai/apertus-v1.5-8b` for all 260 affairs (836 calls). The 70B model was rate-limited (HTTP 429) on the public gateway during the whole hackathon window, so no 70B numbers are reported; the pipeline is model-agnostic and the comparison is one `.env` change.
- **How:** zero-shot structured extraction, JSON output, streaming (required: the gateway cuts non-streamed responses after ~100 s, which a slow 8B generation of a long JSON exceeds).
- **Where:** Public AI OpenAI-compatible endpoint. Cost and throughput in Section 10.
- **No other models** were used, neither for extraction nor as automatic judges. The gold set was corrected by hand against the source documents, as described in Section 6.

6. Data

Sample (`data/sample`, 7 MB, committed): 260 affairs, four intervention types × 18 bodies, begin dates 2022 to March 2026 (recent enough to be representative, old enough to have answers). Languages by detected document language: DE 144, FR 92, IT 24. Levels: federal 16, cantonal 160, municipal 84. Source: OpenParlData API, CC BY 4.0, "Source: OpenParlData.ch"; only material the parliaments themselves published. `docs/data-notes.md` records what the data looks like.

Gold set (`data/gold`): 36 affairs, about two per parliament (one interpellation or question with an answer where available, one motion or postulate), chosen as the *shortest* affairs per body to keep review tractable. Each gold file was pre-filled from the 8B extraction and then corrected by hand against the source documents (review sheets generated by `parliparla review-sheets`, one per affair, listing every candidate value with its quote), with every change logged in the file's `notes`. The number of corrections is itself a result (Section 7).

7. Evaluation

Accuracy on the 36 reviewed gold affairs (`parliparla evaluate`, `data/results/metrics_apertus-v1.5-8b.json`):

Field	Metric	8B	n
Affair type	accuracy	83 %	36
Submission date	accuracy	94 %	36

Field	Metric	8B	n
Submitters	precision / recall / F1	0.97 / 0.78 / 0.81	36
Items (sub-questions, demands)	precision / recall / F1	0.92 / 0.89 / 0.89	36
Item count exactly right	accuracy	92 %	36
Answer present or absent	accuracy	97 %	36
Answer author / answer date	accuracy	90 % / 90 %	20
Answer part → item alignment	precision / recall / F1	0.70 / 0.70 / 0.70	20
Recommendation	accuracy	95 %	20
Decision outcome	accuracy	75 %	8

Correction rate. 11 of 36 gold files needed no substantive correction (the pre-filled "recommendation: null" in affairs without an answer is a template artefact, not a model error, and is not counted). The 25 corrected files show six recurring error classes:

1. **Co-signatories dropped** (8 affairs): the model lists the first signatory or the group and omits the 7–34 co-signers printed on the form; in 4 cases all names were concatenated into one string. Hence recall 0.78 on submitters while precision is 0.97.
2. **Type from register cues instead of the document heading** (6): a federal *Anfrage* labelled interpellation, a federal *Motion* labelled postulate, a *Fraktionsanfrage* and an *Interrogazione* labelled interpellation, a Nidau interpellation labelled motion.
3. **Alignment granularity** (5): when the executive answers several questions in one block ("zu Fragen 4 bis 7", or one undivided preavviso), the model fabricates one part per question; when the executive declines to comment, it emits an empty link list. Both pull alignment F1 down to 0.70 although the items themselves are right.
4. **Dates from the wrong place** (3): PDF signature timestamp instead of the printed date; answer date used as submission date.
5. **Items from the wrong section** (4): proposals quoted from the justification instead of the formal "Conclusion"; a missed opening demand sentence in a federal motion.
6. **Invented presence** (3): a decision "answered" asserted from the executive's answer date when no parliamentary decision document exists; an "answer" extracted from a placeholder PDF ("Antworten werden nicht übersetzt"); a response built from a stray drafting note.

Two further gold files document *data* defects rather than model errors: Vaud's attached documents for two affairs belong to a different affair (a Geneva motion; a 2019 answer to an unrelated resolution).

What this means for the schema question: the item layer is solid (F1 0.89, count exact in 92 %); the answer-alignment layer works but needs a "one block answers several items" rule in the prompt and a fixed vocabulary for "declined" links; co-signatory extraction needs a dedicated pass over the signature block. None of the error classes is specific to a parliament.

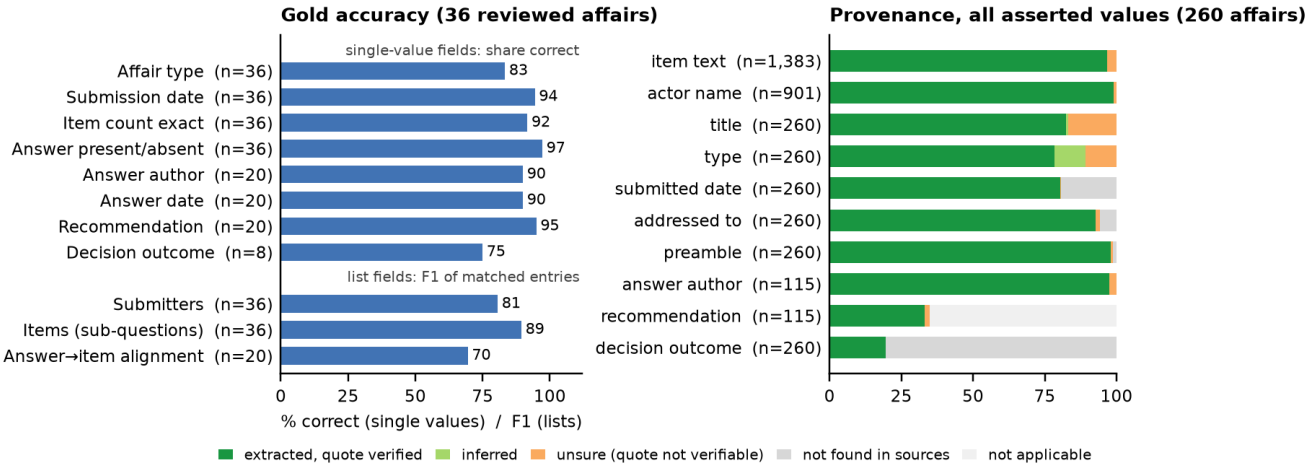


Figure 1. Left: 8B extraction against the 36 hand-corrected gold affairs. Single-value fields (type, date, author, …) are scored as the share of affairs where the value is exactly right; list fields (submitters, items, answer-to-item links) as F1 over matched entries, since a list can be partly right. n = affairs in which the field applies. Right: provenance status of every value the model asserted over all 260 affairs: 4,014 values (54 %) carry a quote that code located in the source, 352 (4.7 %) are marked unsure because the quote was not found, 2,969 fields were not in the attached documents.

Per field, the share of *asserted* values that verified: items 97 % (1,337/1,383), preamble 99 %, addressed body 98 %, submitted date 99.5 %, title 83 %, type 88 % (the 8B model often quotes a paraphrase of the type word), actors 83 % (co-signatory lists with line breaks).

Items per affair: interpellations median 5, written questions 4, motions 2, postulates 1.5; only 10 of 260 affairs yielded zero items.

8. Findings: is a common schema feasible?

	Submitters	Date	Items	Response	Alignment	Decision	
CHE Federal Assembly	19 %[1]	0 %[1]	100 %	100 %	88 %	0 %	federal
ZH Zürich	100 %	100 %	100 %	75 %	62 %	25 %	cantonal
SG St. Gallen	100 %	100 %	100 %	94 %	69 %	0 %	
GR Graubünden	100 %	100 %	100 %	100 %	100 %	50 %	
BS Basel-Stadt	100 %[2]	6 %	94 %	31 %	25 %	94 %	
BE Bern	100 %	100 %	100 %	6 %[3]	6 %	6 %	
LU Luzern	100 %	100 %	100 %	8 %	8 %	0 %	
JU Jura	100 %	94 %	100 %	25 %	25 %	25 %	
NE Neuchâtel	100 %	100 %	100 %	0 %	-	0 %	
VS Valais	75 %	81 %	100 %	44 %	38 %	12 %	
TI Ticino	100 %	100 %	100 %	8 %	8 %	0 %	
VD Vaud	62 %[4]	44 %	44 %	44 %	38 %	12 %	
Stadt Zürich	94 %	94 %	94 %	69 %	50 %	75 %	municipal
Stadt Bern	100 %	100 %	100 %	44 %	38 %	12 %	
Lugano	92 %	92 %	92 %	42 %	33 %	0 %	
Lausanne	88 %	94 %	94 %	19 %	12 %	31 %	
Ville de Genève	100 %	100 %	100 %	0 %	-	0 %	
Nidau	83 %	67 %	100 %	100 %	83 %	0 %	

Figure 2. Share of affairs per parliament with a verified value in each schema layer (8B, n = 260; "–" = not applicable). Left three columns: the intervention layer. Right three: the response and decision layers. [1] Federal texts are body text only; title, author and date live in the register (status `inferred`). [2] Basel-Stadt intervention PDFs carry no submission date in their text (the register has it). [3] Bern attaches one ~90,000-character "Beilage Anfragentexte" per session, not per affair. [4] Vaud attaches external web pages and audio instead of its own documents.

What holds everywhere. The intervention layer: items with verbatim text (≥ 92 % in 17 of 18 bodies, all but Vaud), submitters, date and addressed body. This layer is the same object in Bern, Lugano and the Federal Assembly, and a municipality's motion fits the same five fields as a federal interpellation. **Schema verdict: feasible.**

What holds only where published. The response layer is present for 16/16 federal, 61/160 cantonal and 38/84 municipal affairs. A keyword scan of all 164 affairs without an extracted response found an answer marker in only 2 (Bern city 2026.SR.0095, Bern canton 2025.GRPARL.551), i.e. in 99 % of cases the answer is simply not among the attached documents. Three publication patterns cause this: (a) only the intervention PDF is attached (Geneva city, Jura, Lucerne, Neuchâtel, Ticino, St. Gallen for some types); (b) the executive's statement lives inside the council minutes (Zurich city: "Protokollauszug substanziell", 38k characters); (c) batch documents per session (Bern canton). **Schema verdict: feasible, extraction blocked by data, fixable on the OpenParlData side by attaching the answer documents the parliaments do publish.**

What differs structurally. Decisions: 51 of 260 affairs carry a decision with a verified quote; most parliaments do not attach minutes or decision extracts, and where they do, the decision is a one-line "Das Geschäft ist erledigt" inside 40,000 characters of debate. Recommendations exist only for motions/postulates (38 verified; 23 reject, 8 convert to postulate, 3 accept, 6 partial). Answer stance distribution over 402 answer parts: answered 271, partially 66, declined 46, deferred 19 — the executive declines or defers 16 % of the questions it is asked.

Where the 8B model is weak. Type words quoted as paraphrases (12 % unsure), co-signatory lists with line breaks (17 % of actor names unsure), and a tendency to align every answer part to *some* item (coverage close to 100 % in most parliaments; the gold review in Section 7 measures how much of that is right).

9. One cross-parliamentary question

"How many of the questions in an interpellation does the government actually address?" From the extracted alignments (interpellations and written questions with an answer, n = 70):

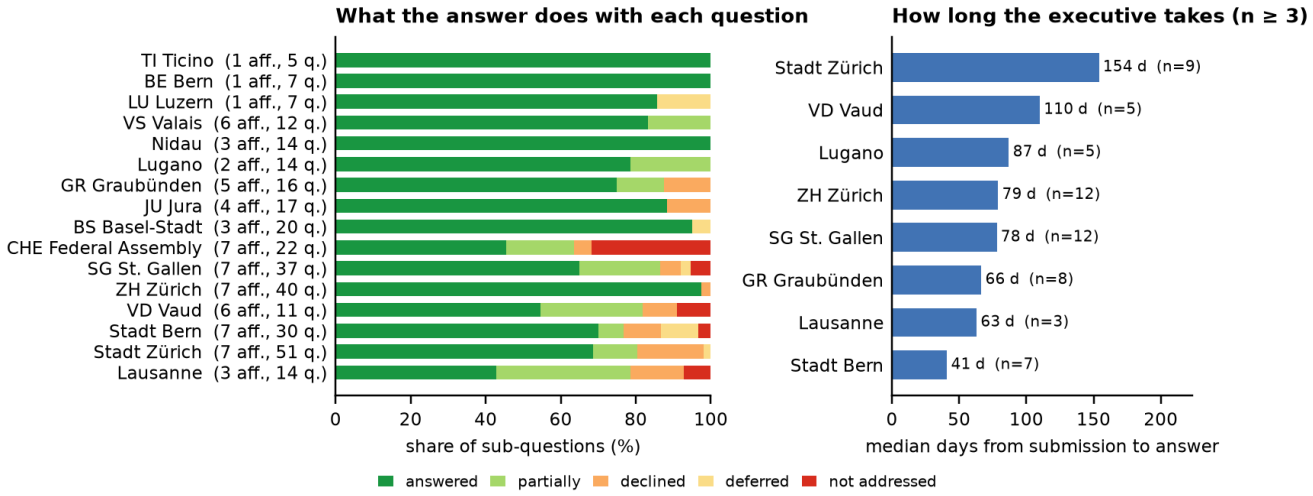


Figure 3. Left: for every sub-question in an answered interpellation or written question, what the executive's answer does with it (n = 70 affairs, 317 sub-questions). Right: median days from submission to the executive's answer, from the extracted dates, for parliaments with at least 3 affairs carrying both dates. Bern, Lucerne, Ticino and Valais have 1–2 such affairs and are omitted; Jura is omitted because its answer documents repeat the question's date in 3 of 4 cases.

Neither number exists in the register today; both come straight from the schema. The 100 % bars should be read with Section 7 in mind: the 8B model over-aligns, so the true figure is a lower bound of the gold-corrected one.

10. Cost and sovereignty

	Apertus v1.5 8B, Public AI
Affairs	260
Model calls	836 (3.2 per affair)
Prompt / completion tokens	1,179,119 / 264,102 (median 3,558 prompt tokens per affair)
Cost	USD 0.17 total → **USD 0.66 per 1,000 affairs** → ≈ USD 215 for the whole 326k corpus
Wall clock	75 s per affair single-stream; 35 min for 260 with 10 parallel workers

Everything runs through an OpenAI-compatible endpoint, so the same pipeline runs on-premise with llama.cpp or vLLM. The text-only 8B weights fit a 12 GB consumer GPU at Q8 (8.6 GB) and a 16 GB CPU box at Q4 — within what OpenParlData's 20 GB VPS can host.

11. Limitations

- **Page numbers.** OpenParlData's served text has no page markers; provenance is a character span, not a page. Per-page re-extraction from the PDF (`url_oparl`) is a straightforward add-on.
- **Sample size.** 260 affairs is a time-boxed *breadth* sample (every parliament, every type), not a depth sample: per-parliament figures rest on 12–16 affairs, and where a parliament rarely attaches answers, response-layer statistics rest on a handful. At USD 0.66 per 1,000 affairs the full 326k corpus is a one-day, USD 215 job with the same pipeline.
- **Gold set size and origin.** 36 affairs, corrected by one reviewer, not an independent double-annotated set. Reported accuracies are indicative, not definitive.
- **8B over-alignment.** The model rarely leaves an answer part unlinked; stance labels are plausible but not validated beyond the gold subset.
- **No 70B comparison** (gateway rate limit), no local-hardware timing.
- **Sample bias.** Gold affairs are the shortest per body; the full sample skews to recent years.
- **Long minutes** are head+tail truncated at 28,000 characters; decisions buried mid-document can be missed.

12. Reproducibility

`make run` builds the Docker image and runs check → extract → evaluate → analyze on the committed sample (default: the first 40 affairs, ≈ 6 minutes with 8 workers; `parliparla run --limit 0` for all 260). Set `LLM_NAME`, `LLM_BASE_URL`, `LLM_API_KEY` in `.env`. Temperature 0, seed 42; the hosted gateway is not bit-reproducible but field-level results are stable. The full 260-affair results and analysis used in this report are committed under `data/results/`. The submitted state is the git tag `submission` in the repository.

13. Next steps

1. Attach answer documents on the OpenParlData side for the parliaments that publish them (the single biggest lever: it turns the response layer from 38 % to near-complete).
2. Page-level provenance from the PDFs; a reviewer UI diff view in the extraction workbench.
3. Split batch documents (Bern canton) and minutes (Zurich city) into per-affair sections before extraction.
4. 70B and local-GPU runs; a 200-affair double-annotated gold set with inter-annotator agreement.
5. Submit PAS v0.1 to the eCH "Politische Geschäfte" working group as a candidate content layer.

License

Code Apache-2.0; this report and the docs CC BY 4.0; sample data CC BY 4.0 (OpenParlData.ch).

References

- OpenParlData API: https://api.openparldata.ch/documentation
- Apertus v1.5: https://huggingface.co/swiss-ai/Apertus-v1.5-8B ; technical report arXiv:2610.15542
- Akoma Ntoso (OASIS LegalDocML), Popolo, OParl 1.1, Parla-CLARIN
- eCH Fachgruppe Politische Geschäfte: https://www.ech.ch/de/der-verein/fachgruppen/politische-geschaefte